

Open-Networking Deployment: Feedback and Trends

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Product Line/Solutions, Edgecore Networks



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Deployment Use Cases Covered

- AI/ML GPU Backend Fabric
- DC Multi-Tenant VXLAN/EVPN Overlay
- Branch-Office/Factory Edge Switch

- All deployments with Edgecore Open Networking Switching/Routing hardware platforms
- All deployments have used Community or Licensed SONIC
- All deployments have used combination of existing transceivers, interconnects on site or as available from the open vendors

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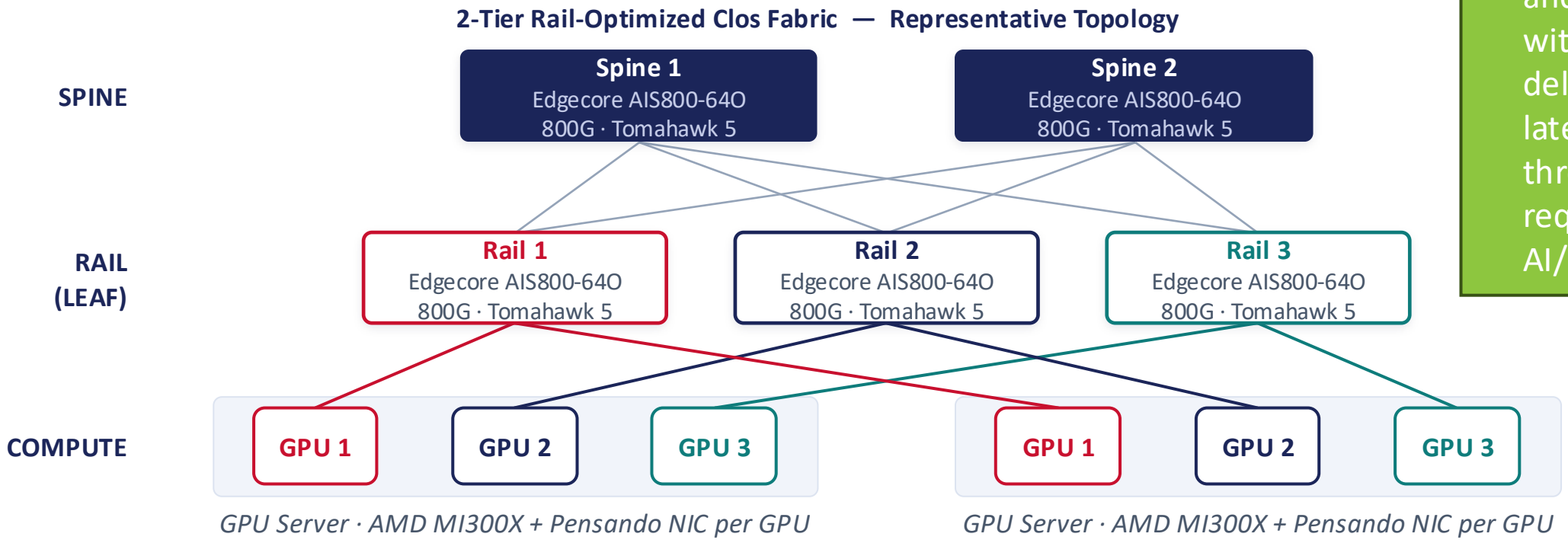
AI/ML GPU Backend Fabric

AMD GPU Cloud · RoCEv2 Ethernet Fabric · Rail-Optimized Clos

- Edgecore AIS800-640
- Supported SONIC
- AMD GPU + Pensando NIC

- ROCEv2 with PFC/ECN and in combination with DLB was able to deliver lossless, low-latency, high-throughput requirement for AI/ML fabric

MI300X → MI355	**RoCEv2**	**800G / 51.2 Tbps**
AMD Instinct GPUs	AI Fabric Protocol	Edgecore AIS800-64O



Rail 1 Rail 2 Rail 3

Each GPU's NIC maps to a dedicated rail (leaf) across every server in the pod — enabling non-blocking, any-to-any GPU communication over lossless RoCEv2 Ethernet (PFC + ECN).

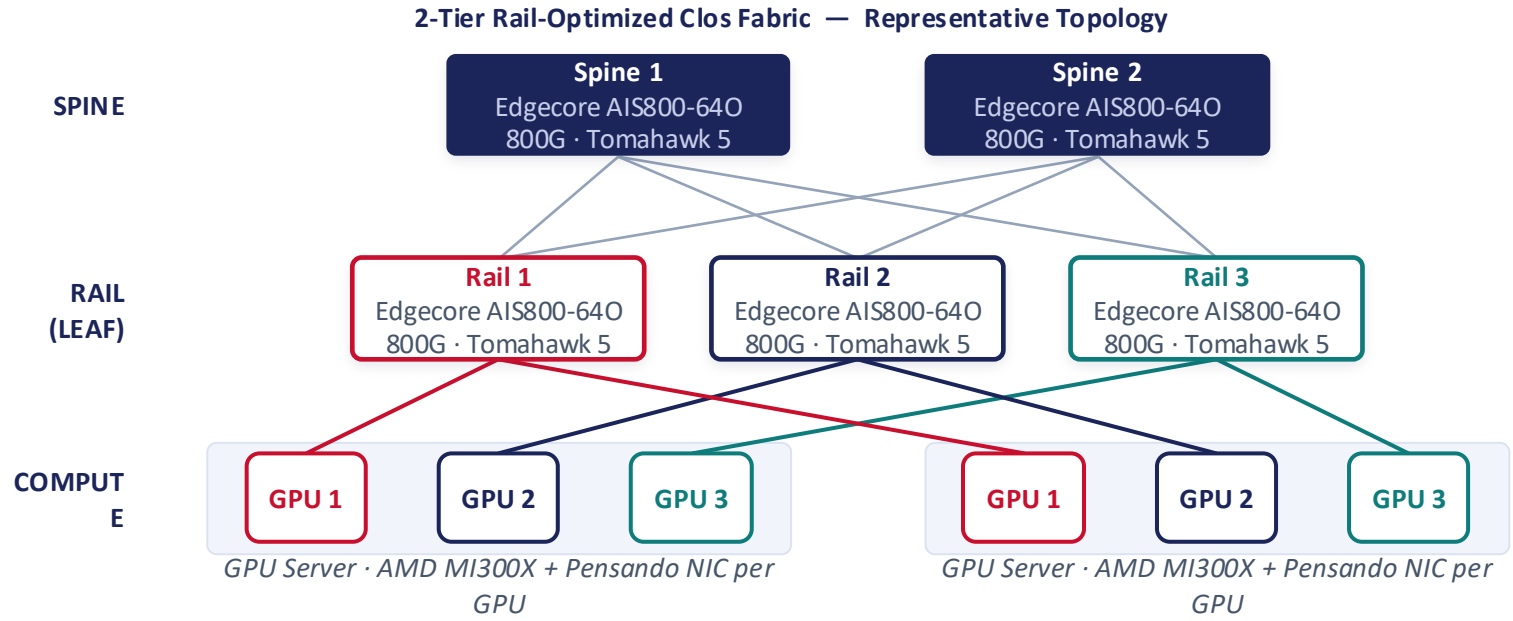
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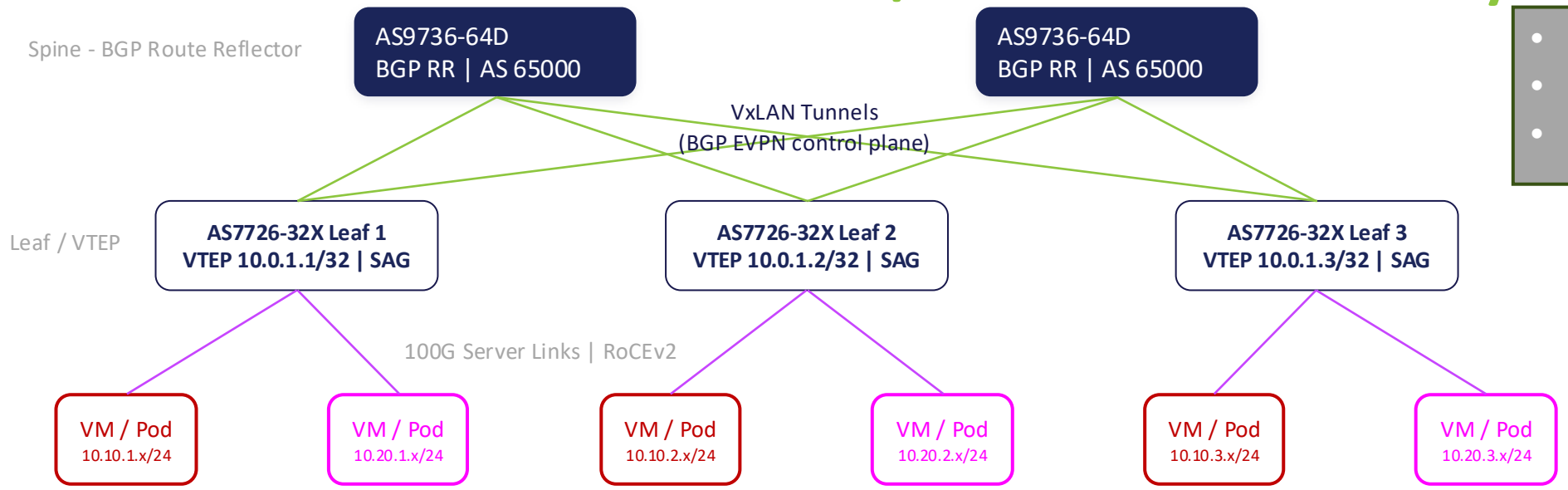
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- Compatibility between GPU NICS, Transceiver, Cables, Switch NIC:
  - Link-level stability with Multi-vendor combination
  - NIC & Switches mismatch on DSCP mapping, MTU or PFC priority
  - ✓ Use Qualified Optics Only
  - ✓ Push one QOS template for both NICS through automation
- Job Completion Times Increase:
  - Regular Counters look normal
  - ✓ Look into ECN, PFC counters and baseline xCCL collectives before handover

DC Multi-Tenant EVPN/VXLAN Overlay



- Edgecore AS9736-64D
- Edgecore AS7726-32X
- Supported SONIC

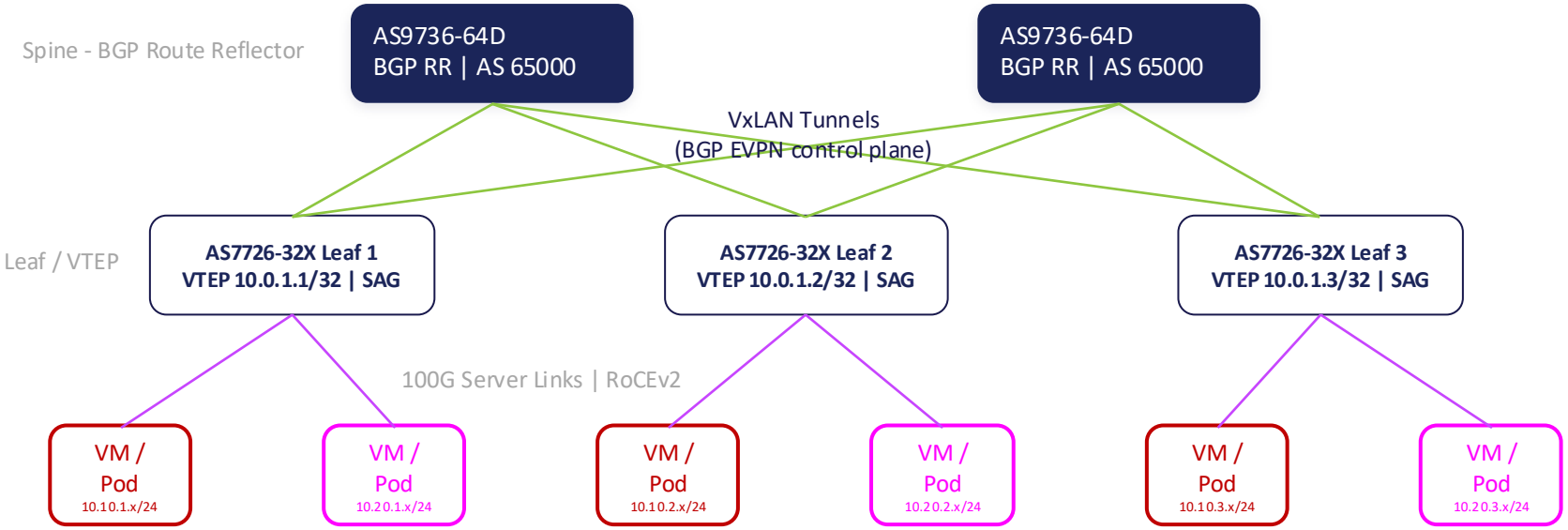
- Basic L2/L3 EVPN/VXLAN Supported in Community SONIC

Tenant A - VRF-A | L3VNI 10100 | VLAN 10 | VNI 10010

Tenant B - VRF-B | L3VNI 20100 | VLAN 20 | VNI 20020

EVPN/VxLAN Overlay	Tenant Isolation	Distributed GW & HA
✓ BGP EVPN	✓ VRF per tenant (L3)	✓ SAG anycast gateway
✓ Type 2 MAC/IP routes	✓ L2VNI per VLAN (L2)	✓ MCLAG active-active ToR
✓ Type 5 IP prefix routes	✓ L3VNI per VRF	✓ DAI + DHCP Snooping
✓ L2 VxLAN + L3 VxLAN	✓ VRF route leaking	
✓ ND/ARP suppression		

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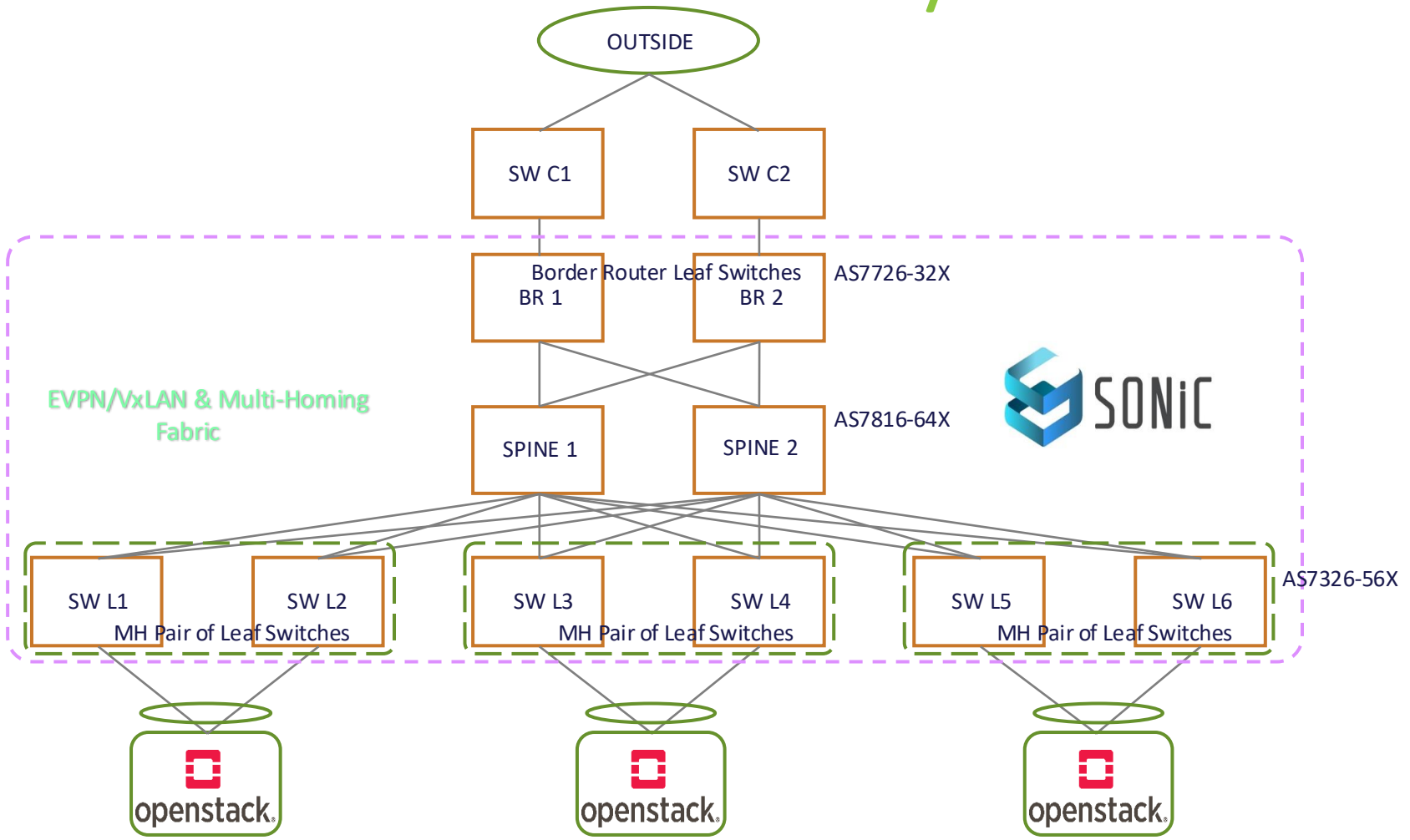
- Edgecore AS9736-64D
- Edgecore AS7726-32X
- Supported SONIC

- Several Feature Combinations Missing
  - DHCP Relay over VXLAN
  - MC-LAG over VXLAN
  - QOS across VXLAN Fabric
  - ✓ Supported and Verified SONIC a better option for deployment

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DC Multi-Tenant EVPN/VXLAN Overlay

- Edgecore AS7326-56X
- Edgecore AS7816-48X
- Edgecore AS7726-32X
- Supported SONIC

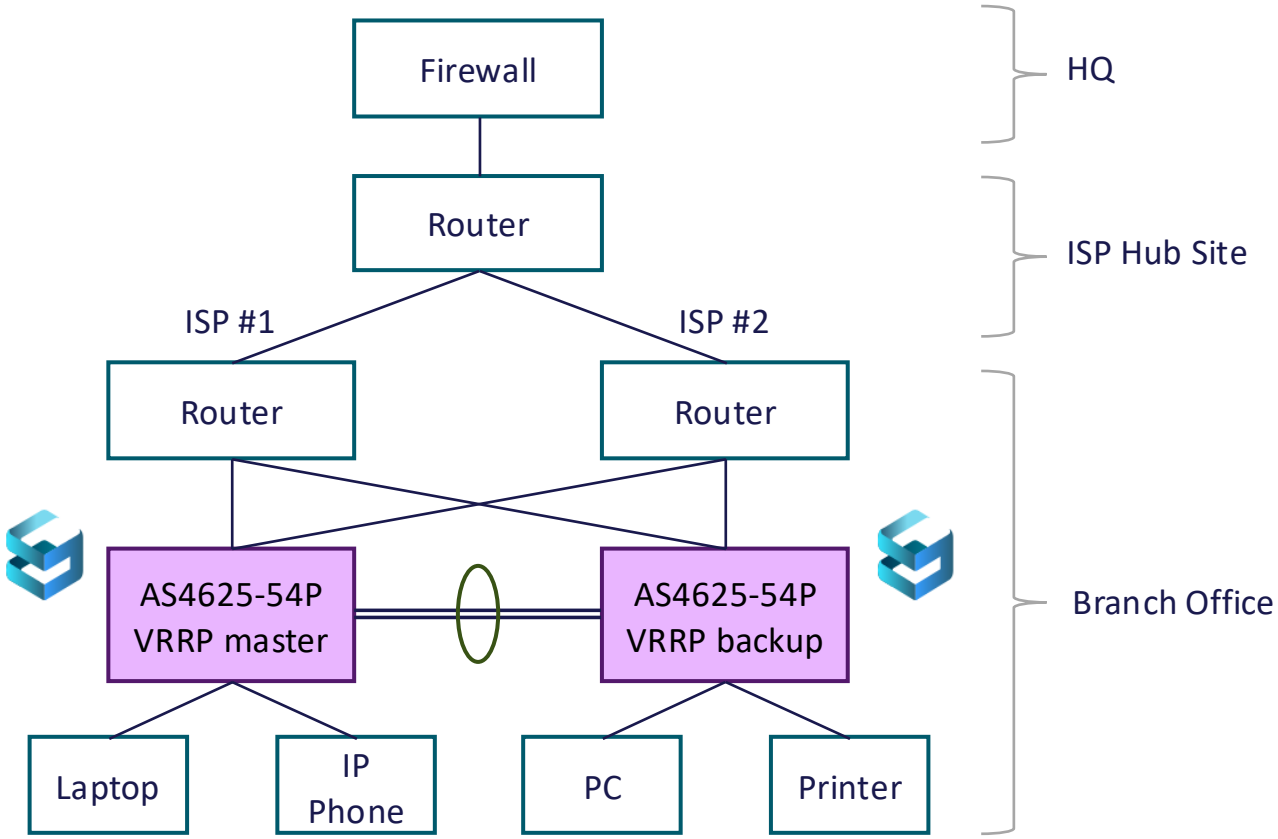


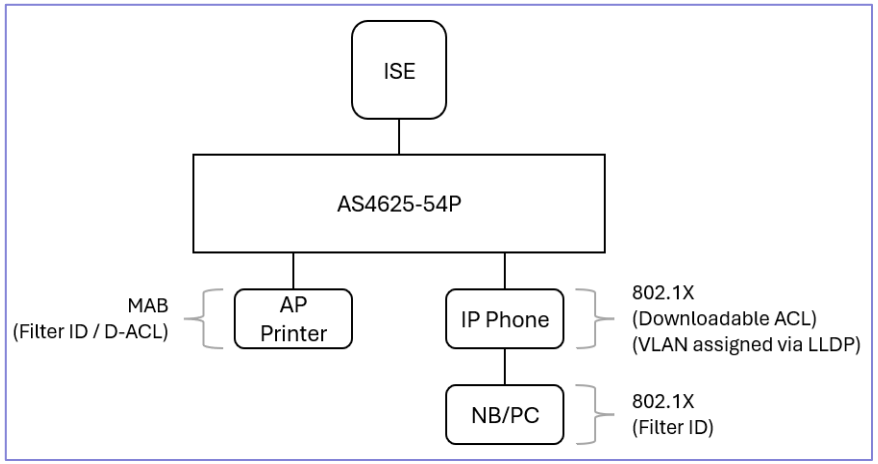
- EVPN/VXLAN Fabric with MH ESI-Based, all-active
  - MAC Nbr sync across MH peers not consistent during age-out, re-learning or VM movements, leading to connectivity loss
  - Fixes requires coordinated testing with SONIC fixes.
  - ✓ VM Lifecycle events (boot, live migration, IP(re-assignment)) full life-cycle requires to be tested.
  - ✓ Supported and Verified SONIC a better option

Branch Office – Factory Edge Switch

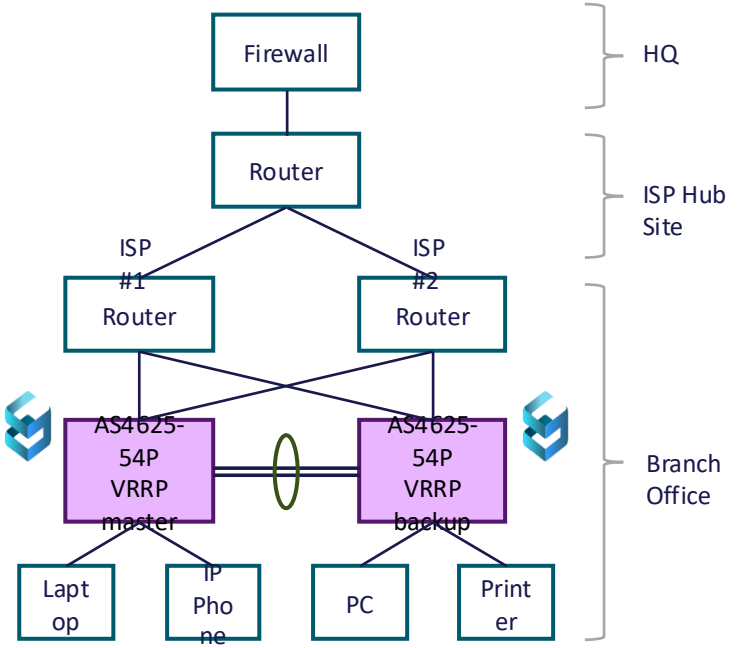
- Edgecore AS4625-54P
- Supported SONIC
- Hub site: VRRP, Dual ISP, Dual Routers, Firewall, ISE(RADIUS/CoA), TACACS+
- Branch:802.1X(NB/PC),802.1x + LLDP Voice VLAN, MAB (Printer, AP) – PoE+ Powered

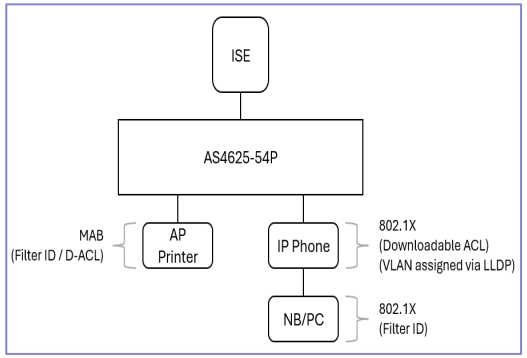
- Network Services works well in Hubsite





Branch Office – Factory Edge Switch





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- Interop / Unsupported Features
  - Diverse End-points required different authentication methods
  - Community SONIC lacked breadth of AAA capabilities such as dynamic VLAN and ACL assignment when interop with ISE server
  - Fixes requires both feature enhancement in SONIC and modifications to ISE configuration
  - ✓ Campus Edge deployments require a far richer AAA feature set than available in Community SONIC.

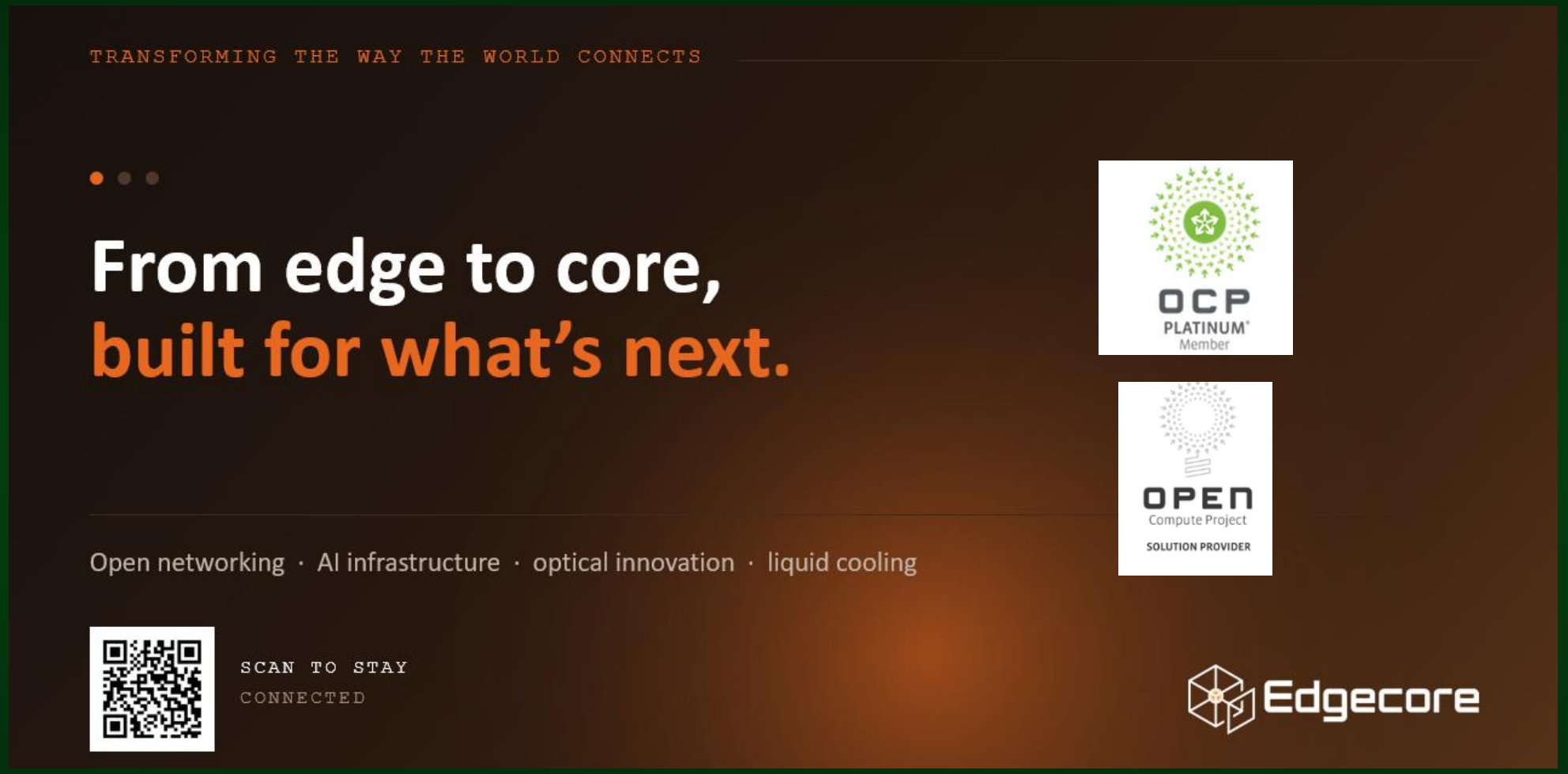
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Requirement for Verified Solutions

- Problem to solve
  - Important for community to publish “Implementation Reports”
  - Important to highlight licensed/supported SONIC feature sets used for production, for varied industry adoption

- Where to find additional information
  - [Open Networking Product Portfolio] https://www.edge-core.com/products/
  - [Open Networking Deployment Use Cases] https://www.edge-core.com/resource-centers/

Thank You!


TRANSFORMING THE WAY THE WORLD CONNECTS
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